

Business Document Ingestion

Client Output, Reporting, and CRM Access Overview

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Client Output, Reporting, and CRM Access Overview

Read this first

This guide explains what a completed document-processing engagement can give you, what each output is for, and how approved data can later be searched, reported on, exported, or prepared for a CRM. It is written for business users; you do not need to know the internal processing steps to use the results.

This edition also covers the optional visual-intake pilot and source-only handoff implemented in this checkout. It does not claim a deployed client service or a completed merge. The stable release remains 1.0.0; current repository and verification state is recorded separately in `HANDOFF.md`.

The important distinction is simple:

If the final review is...	You receive...	What it means
Open or blocked	A review package and an exception register	Questions still need an answer. The data is useful for review, but it is not released as approved operational facts.
Clear	An approved data package, reporting/retrieval snapshot, and optional CRM-ready files	Every included fact is provenance-linked, review-clear, and tied to the exact source snapshot.

The system does not hide an unresolved item to make a spreadsheet look clean. It records the question, reason, source, and next step. A completed pipeline is therefore more than a collection of extracted rows: it is a traceable answer to what was found, what was proven, and what still needs a decision.

The end product at a glance

A client delivery is organized into five practical layers. Depending on the engagement and final gate, some layers may be delivered as blocked findings rather than approved data. That distinction is visible in the package.

1. **Source and evidence layer.** Original documents, page references, hashes, and citations that let a user trace an important value back to its source.
2. **Review and control layer.** The review queue, validation, arithmetic, attribution, completeness, and exception results. This explains what was checked and what was not proven.
3. **Approved data layer.** A canonical JSON export containing only review-clear, provenance-linked facts, plus an exclusion artifact for every withheld record or field.

4. **Reporting and retrieval layer.** A read-only, immutable data snapshot that supports governed search, record lookup, account/contact cards, analysis, standard reports, and bounded exports.
5. **CRM handoff layer.** Checksummed no-send staging files and optional common CRM-import files. These prepare a controlled load; they do not upload, overwrite, or authorize a live CRM change.

The package is versioned by the source-export hash and supporting checksums. If an authorized correction changes a fact, a new snapshot is built. The old snapshot remains available for audit; it is never edited in place.

What the files look like

The exact folder names may be tailored to the engagement, but the contents have the following roles.

Output	Typical form	What you use it for
Final review package	JSON, CSV, XLSX, and HTML	Review open questions, filter by priority or document, and return controlled decisions.
Control results	JSON and human-readable summaries	Understand arithmetic, validation, attribution, completeness, sampling, and final-gate results.
Exception register	JSON and reviewer views	See every withheld record, missing input, failed check, or unresolved value with its reason.
Canonical export	Versioned JSON plus exclusion artifact	The authoritative machine-readable approved-fact handoff. It is the source for downstream loading and retrieval.
Load plan	Versioned, checksummed JSON	See the dependency order, record counts, keys, and integrity checks required before loading another system.
Retrieval snapshot	Immutable SQLite database	The local, read-only source used by the MCP and REST API. It contains approved facts only.
CRM staging package	Ordered CSV files, API envelope, and manifest	Give a target-system team a full, reconciled, no-send handoff across all canonical tables.
Common CRM import package	Twelve ordered CSV files, mapping template, source catalog, and write plan	Start target-CRM mapping and sandbox testing using familiar business objects.
Optional source-intake package	Private original images, proposal history, exceptions, manifest, and a derived PDF when complete	Hand new source pages to the processing team. This is an evidence archive, not an approved CRM import or a report.

The Excel files

Excel is used as a convenient review or handoff view, not as a place where facts silently change.

- The **final review workbook** is for answering open questions. Its issued copy and returned copy are preserved separately. Returned decisions are proposals until an operator reviews the exact impact and reruns affected controls.

- The **schema and CRM capability workbook** is an orientation aid. It explains tables, fields, API/MCP capabilities, sample report shapes, and import-file structure. It does not contain hidden authority to load a CRM or change an approved fact.
- The **CRM import mapping template** is where a target-system owner completes vendor- and tenant-specific object and field mapping. Its included suggestions are starting points, not approved target mappings.

What “approved data” means

An approved row is not simply a row that an AI model read. Before it reaches the canonical export or the retrieval snapshot, the system requires applicable controls to be satisfied separately:

- source and page provenance are retained;
- the record has a review-clear status;
- applicable arithmetic is proved or explicitly not applicable;
- the exhaustive final review queue has no open item for that record; and
- every relevant attribution, completeness, and approval condition is recorded.

If a source, ledger export, payment export, mapping decision, or review answer is unavailable, the package says so. It does not manufacture a substitute fact. This is why a delivery can be valuable even when it is not ready for a CRM load: it makes the outstanding business decisions visible and bounded.

What an MCP is

MCP stands for **Model Context Protocol**. In plain language, it is a controlled connection that lets an AI assistant use approved business tools. Instead of pasting spreadsheets into a chat, the assistant can ask the approved retrieval snapshot specific, bounded questions and show you the result.

Think of it as a librarian with a fixed catalog:

1. You ask your assistant a business question in normal language.
2. The assistant selects an approved tool, such as search, exact lookup, sales analysis, or a standard report.
3. The tool reads one immutable, approved-fact snapshot.
4. The result includes source-export identity, checksums, and relevant source document IDs so it can be checked.

The approved-data MCP does **not** let the assistant run arbitrary SQL, browse raw AI responses, invent records, bypass the review gate, or write to a CRM. Tool results do become part of the conversation context, so the client must approve the selected AI provider’s privacy, retention, residency, and user-access settings before connecting real data.

How you can connect

Choose the connection that fits how your team will work. The same approved snapshot and semantic rules are used by every option; the transport changes, not the facts or calculations.

Your need	Connection	Suitable clients	What is required
A quick, private desktop pilot	Local studio MCP	Claude Code, Claude Desktop, and compatible local desktop clients	The completed run’s local immutable snapshot and a private local registration. This is the recommended first connection.

Your need	Connection	Suitable clients	What is required
A controlled internal application or BI tool	TLS/bearer REST API	An approved internal application or BI consumer	Explicit enablement, TLS, dedicated rotated bearer secret, network controls, monitoring, and client acceptance. This is an API, not an MCP connector.
Web, iOS, or organization workspace use	Remote Streamable HTTP MCP at a public <code>/mcp</code> URL	Claude web/desktop/mobile after connector setup; selected ChatGPT workspace/API surfaces	Public HTTPS, OAuth/OIDC authorization, exact tenant/role/scope rules, approved workspace registration, monitoring, and live acceptance.

Local desktop connection: the simplest first step

After the engagement reaches the approved-data stage, the operator registers a private local MCP configuration that points to the exact local snapshot. In Claude Code, this is normally a local stdio registration; Claude Desktop and other compatible desktop clients use the same launcher through their local MCP configuration. The database stays on the workstation, and the server is read-only.

This is the right starting point when one approved team is working from the machine holding the completed snapshot. It does not make the service public or available on a phone.

Remote connection: needed for mobile and web

Claude mobile, claude.ai, and selected ChatGPT web, desktop, or mobile experiences require a separately deployed remote MCP service. The repository already supplies the remote approved-data server, but a real client deployment still needs:

- one public HTTPS URL ending exactly in `/mcp`;
- a trusted TLS certificate and reverse proxy;
- a real OAuth/OIDC authorization service with token introspection;
- a specific tenant claim, approved roles, and `crm:read/crm:export` scopes;
- approved logging, alerting, retention, backup, export cleanup, and incident response procedures; and
- connector registration and representative acceptance tests in the selected Claude and/or ChatGPT workspace.

Mobile users normally use an already configured connector; they do not create the deployment from the mobile app. Product-plan and workspace availability can change, so the chosen provider surface is tested during the deployment acceptance rather than assumed from protocol compatibility.

What you can ask for

The system is designed for normal business questions. The assistant chooses a bounded tool and returns a small, checkable result; it does not send an entire large report into a chat conversation.

Find a record when you do not know its key

Use search for a company name, invoice number, address fragment, acknowledgement number, job, project, or other source-visible phrase.

Examples:

- “Find documents and records related to Acme Industrial.”
- “Search for invoice INV-1042 and show the approved source citation.”
- “Find the address containing 125 Harbor Drive.”
- “Which records mention ACK-7781 or job 44-19?”

The MCP uses `search_crm_records` for this cross-table discovery. Once the right row is known, `get_crm_record` retrieves the exact canonical record by table and key.

Pull an account, address, or contact card

`get_account_card` is the preferred way to ask for a business-ready view of a company or party. It joins approved party names and roles with addresses, contacts, invoice/payment activity, related source documents, and summary figures.

Examples:

- “Show the contact card for Acme Industrial, including addresses and contacts.”
- “Give me the approved address and invoice history for customer CUST-104.”
- “Which source documents support this vendor’s contact information?”

Money is separated by currency in account cards. The service will not add US dollars, Canadian dollars, euros, or other currencies into one misleading total.

Slice and aggregate sales safely

`analyze_sales` is the governed analysis tool. It can group up to three business dimensions, including company/customer, vendor, region, selling location, month, quarter, year, currency, acknowledgement number, and job number. It can also filter by date, those same dimensions, and invoice-value bounds.

Examples:

- “Show sales by company and region for Q2, separated by currency.”
- “Compare sales by selling location for this year.”
- “Break down customer sales for job J-204 by month.”
- “Which acknowledgement numbers contributed to Midwest region sales?”

Currency is always treated as a grouping dimension, even if it is not mentioned in the question. That prevents unlike currencies from being added together.

Query one approved table

`query_crm_records` is for a focused, bounded slice of one table. It supports approved filters, selected fields, stable sorting, and pagination. It does not execute formulas, code, or arbitrary SQL.

Examples:

- “List approved invoices for Acme Industrial with invoice number, date, amount due, and source document ID.”
- “Show the next 50 approved shipment records, using the available schema fields.”
- “Find addresses whose city starts with ‘New’, then show the account cards for those parties.”

The last example needs more than one tool call: the one-table query does not perform arbitrary joins from contacts to addresses. The assistant must discover available fields and keys before choosing filters, not invent columns.

For a larger reconciliation, `export_crm_table` returns checksummed pages rather than an unbounded conversation response.

Standard reports available out of the box

The retrieval layer supplies seven deterministic reports over approved facts. Each report is bound to the snapshot hash and includes source document IDs for its supporting data.

Report	Typical use
<code>account_directory</code>	List approved accounts/parties with key contact and address information.
<code>invoice_register</code>	Review invoice-level amounts, dates, parties, and source citations.
<code>receivables_aging</code>	Review the current approved balance snapshot by aging bucket as of a required date. It is not a reconstructed historical ledger.
<code>sales_by_customer</code>	Compare approved sales by customer, with currencies kept separate.
<code>sales_by_selling_location</code>	Compare approved sales by sales office, branch, or selling location.
<code>attribution_coverage</code>	See which approved invoice values have a supported attribution and which do not.
<code>shipment_performance</code>	Review approved shipment activity and related operational measures available in the snapshot.

You can ask an assistant for these in ordinary language: “Run the invoice register for April,” “show sales by customer,” or “run receivables aging as of June 30.” The assistant maps the request to the report and must preserve its parameters and snapshot identity in the result.

Downloads and large exports

Chat is useful for concise answers, not for sending thousands of rows into a model conversation. The remote MCP therefore supports controlled CSV and XLSX export jobs.

An authorized user can request one approved table or one standard report in CSV or XLSX form. The server creates a bounded, short-lived download with:

- the snapshot and source-export hash used at the start;
- the requester identity in hashed form;
- the exact table/report and filters requested;
- row, column, and file checksums; and
- an expiry time and owner-bound status lookup.

Spreadsheet-formula-like text is neutralized in exports. A different user cannot inspect another user’s job, a changed file fails verification, and an expired link is refused. These exports create a read-only report artifact; they never update canonical data or write to a CRM.

CRM inputs: what is ready and what still needs a decision

Adding photographs and proposed records first

The optional intake pilot can retain original PNG/JPEG page images and let the connected, vision-capable assistant read them using the pipeline’s published field dictionary. It can preserve an invoice reading or contact-card observation without changing the approved business database. Unknown document families and unfamiliar labels stay explicit, not forced into made-up fields.

1. Agree which images and AI provider may receive the data; the operator enables intake separately from approved reporting.

2. A compatible application uploads the actual original bytes to a named session. Keep its receipt outside the chat. A chat attachment alone does not prove the service received it.
3. The assistant discovers the schema, reads every retained image, and proposes source-visible fields with original readings, page references and uncertainty.
4. The service checks the structure and returns **pending review** or **rejected**, retaining the original and every finding. Neither status is verification of the reading or permission to publish it.
5. Corrections append another proposal. An uncertain network retry uses the same request key; changed content uses a new one. A later proposal never erases an earlier failure or changes an approved record.

For example: “Read this session’s retained invoice images. Propose the invoice number, visible line items and amounts. Keep unreadable fields and unfamiliar labels explicit, and show me the review receipt.” For a contact card, request source-visible observations, not an immediate customer update.

The pilot accepts 1 to 20 pages per session: PNG/JPEG only, at most 10,000,000 original image bytes and 25,000,000 pixels per page. PDF, HEIC, WebP, animations and image URLs are not direct upload types. Preserve unsupported originals separately. Deployment request limits may be smaller than the image limit. No mobile capture app, attachment relay, session search, shared reviewer screen, or automatic record publication is bundled. Actual upload and image reading must be tested in the selected client; a working report connector proves neither.

The source-only handoff

A local operator can export the session into a new private source package. It retains every upload attempt, proposal version and rejection with checksums. A complete usable image set also gets a session-named PDF and an original-to-page map. The operator verifies the package against a separately retained receipt, visually checks the PDF, and starts the normal profiling/intake commands.

The PDF never replaces the originals. Display conversion keeps page order and encoded orientation and shows transparency on white. It is not a rescan, OCR result, or proof of original physical resolution. Missing pages or conversion failures keep the archive blocked; a failed partial PDF is never used. This is a controlled source archive, not automatically a client-approved delivery.

Independent extraction, business controls, authorization and final review still stand between these sources and a new approved snapshot. The assistant’s proposal is not automatically converted into a pipeline vote and cannot change account cards, sales totals, reports or CRM-import files. Full setup, fields, limits and recovery are in Visual Intake.

Approved-data CRM preparation

The pipeline can prepare CRM-ready inputs without sending anything to a CRM. This separation gives the target-system owner a chance to validate mappings, permissions, duplicate behavior, automations, and rollback before production.

Full canonical staging package

The no-send staging package preserves all **28 canonical tables** in dependency order. It includes ordered CSVs, a versioned API-operation envelope, source and load-plan checksums, row counts, idempotency keys, formula-safe display cells, and authoritative compact canonical JSON for every row. It is designed for a receiver or integration team that needs complete reconciliation across the full data model.

Common CRM import package

For teams that work with familiar CRM objects, the optional common package creates twelve ordered CSV files:

1. accounts
2. account roles
3. addresses

4. contacts
5. products
6. sales locations
7. shipments
8. sales transactions
9. sales transaction lines
10. transaction charges
11. payments
12. payment applications

It also includes a field-mapping template, a vendor-source catalog, a write-readiness plan, exact row and package checksums, and complete coverage accounting for all 28 canonical tables. A field not represented by the common profile is explicitly marked as outside the profile rather than silently dropped.

What must happen before a live CRM write

No live Salesforce, HubSpot, Dynamics/Dataverse, Zoho, or other CRM writer is enabled by this package. A target-specific implementation starts only after the client selects the vendor and tenant, supplies current object/property metadata, approves mapping and credentials, and proves the adapter in a sandbox.

The production-write acceptance must show stable external keys, idempotent reruns, parent-before-child loading, explicit rejected-row handling, count and financial reconciliation, relationship checks, before-images/created IDs for rollback, and separate authorization for the exact package hashes. Read-only MCP/API access does not grant `crm:write` authority.

A practical first-use path

For most teams, the safest and quickest sequence is:

1. Review the final package and resolve or explicitly retain any open items.
2. Confirm the exact approved snapshot and its source-export checksum.
3. Start with a private local desktop MCP connection for representative questions and reports.
4. Validate account cards, invoice search, a sales slice, one standard report, and an exported CSV/XLSX against the delivered package.
5. If mobile, web, or multi-user access is needed, deploy the remote `/mcp` service with OAuth and complete workspace-specific acceptance.
6. If CRM loading is needed, complete target mapping and sandbox testing from the no-send package before authorizing a target-specific adapter.

This order keeps a simple reporting pilot separate from a public deployment or a live CRM change. It also gives business users an early way to test whether the approved data answers their questions before the organization commits to a larger integration.

Questions this guide helps answer

- **Can I find an address or contact quickly?** Yes, use search and the account card over the approved snapshot.
- **Can I see sales by company, region, office, customer, job, or time period?** Yes, use governed sales analysis or the relevant standard report; currency is kept separate.
- **Can I pull any approved record?** Yes, search when the key is unknown, then retrieve the exact canonical record by table and key.
- **Can I download a report?** Yes, a remote deployment can create bounded, owner-bound CSV/XLSX jobs with checksums and expiry.
- **Can an assistant change my CRM?** No. Approved records remain read-only. Optional intake writes proposals to a separate journal, not canonical or target-CRM records. CRM inputs remain no-send until separate adapter implementation, authorization and acceptance.

- **Can I submit a photo for a new record?** The optional pilot supports original PNG/JPEG upload and source-cited proposals through a compatible application. It stops at review, not publication.
- **Can I ask for any possible analysis?** Shipped filters, account cards, sales dimensions and seven reports work within their limits. Custom joins, saved reports, forecasting and broader metrics need additional implementation and appropriate approved source data.
- **Can the system tell me what it cannot prove?** Yes. Unresolved source, arithmetic, attribution, completeness, mapping, and review findings remain explicit rather than being hidden.

Before connecting real data

Use the following checklist with the engagement owner:

- ☐ The final review and applicable business gates are complete for the exact snapshot being served.
- ☐ The team has chosen local desktop, internal REST, or remote MCP access.
- ☐ The selected AI provider's privacy, retention, residency, and user-access terms are approved for the data returned in conversations.
- ☐ Remote use has an approved domain, TLS, OAuth/OIDC, tenant/role/scope policy, monitoring, retention, and incident process.
- ☐ The exact Claude and/or ChatGPT workspace connection has been tested with representative questions.
- ☐ Any CRM load has a completed tenant mapping, sandbox evidence, reconciliation plan, rollback plan, and separate production authorization.
- ☐ Optional intake has separately demonstrated byte upload, image reading, ownership checks, rejection/retry receipts, source export and pending-review wording; none is inferred from successful approved-data queries.

Where to go next

- For the detailed output contract and approved retrieval rules, see Canonical Deployment, CRM Load, and Retrieval.
- For activation, connection, deployment, and acceptance steps, see MCP Production Integration.
- For the target-system import and write-readiness boundary, see CRM Write Readiness.
- For unimplemented capabilities, see the vendor-neutral roadmap.
- For the broader evidence and review workflow, see the Client Overview and Client User Guide.

The guiding rule is unchanged across every output: use the approved snapshot for business answers, keep its citations and checksums with important results, and treat a later correction as a new, controlled version rather than an edit to history.